

Tabular Planning: Policy Iteration

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CS 1840 Fall 2026**

Announcements

HW1 is out!

Check the course website for Office Hours:

<https://harvard-cs-1840.github.io/2026/>

Recap: Bellman Optimality

$$\mathcal{M} = \{S, A, P, r, \gamma\}$$

$$P : S \times A \mapsto \Delta(S), \quad r : S \times A \rightarrow [0,1], \quad \gamma \in [0,1)$$

$$\text{Policy } \pi : S \mapsto A$$

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Bellman Optimality—the Q version

$$Q^*(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left[\max_{a' \in A} Q^*(s', a') \right]$$

Finishing Up From Last Lecture



1: Bellman Optimality

2: An Iterative Algorithm: Value Iteration

Value Iteration Algorithm:

1. Initialization: Q^0

2. Iterate until convergence: $Q^{t+1} \Leftarrow \mathcal{T} Q^t$

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We hope $Q^t \rightarrow Q^*$, as $t \rightarrow \infty$

Convergence of Value Iteration:

Lemma [contraction]: Given any Q, Q' , we have:

$$\|\mathcal{T}Q - \mathcal{T}Q'\|_{\infty} \leq \gamma \|Q - Q'\|_{\infty}$$

Proof:

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$$|(\mathcal{T}Q)(s, a) - (\mathcal{T}Q')(s, a)| = \left| r(s, a) + \gamma \mathbb{E}_{s' \sim P(s, a)} \max_{a'} Q(s', a') - \left(r(s, a) + \gamma \mathbb{E}_{s' \sim P(s, a)} \max_{a'} Q'(s', a') \right) \right|$$

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$$\dots \leq \gamma^{t+1} \|\widehat{Q}^0 - Q^\star\|_\infty$$

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(a fix point iteration alg):

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2. We could set $\pi^t(s) = \arg \max_a Q^t(s, a)$, does $\pi^t \rightarrow \pi^*$ when t increases?

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3. Can we still hope π^t being a good policy?

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Error in Q is amplified by $1/(1 - \gamma)$

(Because π^t could disagree w/ $\pi^{\star}$ at every step)

Question for Today:

Given an MDP $\mathcal{M} = (S, A, P, r, \gamma)$, How to directly search for $\pi^\star : S \mapsto A$

Outline:

1: An Iterative Algorithm: Policy Iteration

2: Convergence? How fast?

3: A new model: Finite horizon MDP

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Policy Improvement $\pi^{t+1}(s) := \arg \max_a Q^{\pi^t}(s, a), \forall s$

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Key properties of Policy Iterations:

1. Monotonic improvement:

$$Q^{\pi^{t+1}}(s, a) \geq Q^{\pi^t}(s, a), \forall s, a$$

2. When $\pi^{t+1} = \pi^t$, then π^t is equal to $\pi^\star$

Monotonic Improvement

Recall: Policy Improvement $\pi^{t+1}(s) = \arg \max_a Q^{\pi^t}(s, a), \forall s$

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$$V^{\pi^{t+1}}(s) \geq V^{\pi^t}(s), \forall s, ??$$

Summary of Policy Iteration

Iterate between Policy Evaluation and Policy Improvement:

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3: A new model: Finite horizon MDP

Finite horizon Markov Decision Process

$$\mathcal{M} = \{S, A, r, P, H, \mu_0\},$$
$$r : S \times A \mapsto [0,1], H \in \mathbb{N}^+, P : S \times A \mapsto \Delta(S), s_0 \sim \mu_0$$

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Very common in control,
e.g., keep tracking a pre-specified trajectory with fixed length and fixed initial state

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Note that in finite horizon setting, we will consider **time-dependent policies**, i.e.,

$$\pi := \{\pi_0, \pi_1, \dots, \pi_{H-1}\}, \pi_h : S \mapsto A, \forall h$$

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Policy interacts with the MDP as follows:

$$\tau = \{s_0, a_0, s_1, a_1, \dots, s_{H-1}, a_{H-1}\}, s_0 \sim \mu_0, a_0 = \pi_0(s_0), s_1 \sim P(\cdot | s_0, a_0), a_1 = \pi_1(s_1), \dots$$

V/Q functions in Finite horizon MDP

$$V_h^\pi(s) = \mathbb{E} \left[\sum_{\tau=h}^{H-1} r(s_\tau, a_\tau) \mid s_h = s, \pi \right]$$

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$$Q_h^\pi(s, a) = \mathbb{E} \left[\sum_{\tau=h}^{H-1} r(s_\tau, a_\tau) \mid (s_h, a_h) = (s, a), a_\tau \sim \pi \text{ for } \tau > h \right]$$

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(i.e., we know how to perform optimally starting at $h + 1$)

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Summary so far

1. Basics of MDPs (e.g., Bellman equation, Bellman optimality)
2. How to perform policy evaluation and how to compute optimal policies

Summary

Attendance

<https://bit.ly/46pCUdN?r=qr>



Feedback

<https://bit.ly/4gYn2nC>

